Reference solution

At along the tangent to an electron orbit, an observer sees radiation only briefly. This is the arrival-time interval of light emitted as the velocity direction sweeps the cone angle .
The electron travels the following distance during that interval:

Let . The interval between emission at the two points is

For the tangent observer, subtract the travel time of light from to from this interval, as given below:

Then

Also,

In the limit ,

Substitution into gives

The numerical data give